IDENTIFYING THE TERMS OF THE HOM(DELTA[1], L) COSKELETON DIAGRAM

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 1995-2054 of sources/stacks-hypercovering/pinned-original-source.tex; proof span is lines 2056-2072, the complete author proof (15 body lines).

Editorial note. Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 20 of queue_hypercovering.json (pb=745, stmt 1995-2054, proof 2056-2072).

Editorial note. stacks-hypercovering wave-1 (hc01): fresh adjudication accept (2026-10-14); pool per standing fallback order (hypercovering -> pic -> spaces-more-cohomology) after md01 closed the moduli pool; hypercovering.tex 53 labels/34 with proof judged 26 accept / 8 reject / 14 pb=0 env-dup rows excluded; spans cite-free and reference/slogan/history-free so no patchenv needed. Named omission: The intersection identity (partial Delta[n+1] coprod partial Delta[n+1] = the intersection inside Delta[1] x Delta[n+1]) is 'left to the reader' - a named mechanical check; the four identifications and the pushout argument are in-span.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 018V: sources/stacks-hypercovering/pinned-original-source.tex, source lines 2072–2072. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 017L: sources/stacks-hypercovering/pinned-original-source.tex, source lines 2072–2072. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 017N: sources/stacks-hypercovering/pinned-original-source.tex, source lines 2072–2072. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:hypercovering.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let $\mathcal{C}$ be a site with fibre products. Let X be an object of $\mathcal{C}$. Let L be a simplicial object of $SR(\mathcal{C}, X)$. Let $n \geq 0$. Consider the commutative diagram*

$$\begin{array}{ccc} \mathrm{Hom}(\Delta[1], L)_{n+1} & \longrightarrow & (cosk_n sk_n \mathrm{Hom}(\Delta[1], L))_{n+1} \\ \downarrow & & \downarrow \\ (L \times L)_{n+1} & \longrightarrow & (cosk_n sk_n (L \times L))_{n+1} \end{array} \tag{0.1.1}$$

coming from the morphism defined above. We can identify the terms in this diagram as follows, where $\partial\Delta[n+1] = i_{n!} sk_n \Delta[n+1]$ is the n-skeleton of the $(n+1)$-simplex:

$$\begin{aligned} \mathrm{Hom}(\Delta[1], L)_{n+1} &= \mathrm{Hom}(\Delta[1] \times \Delta[n+1], L)_0 \\ (cosk_n sk_n \mathrm{Hom}(\Delta[1], L))_{n+1} &= \mathrm{Hom}(\Delta[1] \times \partial\Delta[n+1], L)_0 \\ (L \times L)_{n+1} &= \mathrm{Hom}((\Delta[n+1] \amalg \Delta[n+1], L)_0 \\ (cosk_n sk_n (L \times L))_{n+1} &= \mathrm{Hom}(\partial\Delta[n+1] \amalg \partial\Delta[n+1], L)_0 \end{aligned}$$

and the morphism between these objects of $SR(\mathcal{C}, X)$ come from the commutative diagram of simplicial sets

$$\begin{array}{ccc} \Delta[1] \times \Delta[n+1] & \longleftarrow & \Delta[1] \times \partial\Delta[n+1] \\ \uparrow & & \uparrow \\ \Delta[n+1] \amalg \Delta[n+1] & \longleftarrow & \partial\Delta[n+1] \amalg \partial\Delta[n+1] \end{array} \tag{0.1.2}$$

Moreover the fibre product of the bottom arrow and the right arrow in (0.1.1) is equal to

$$\mathrm{Hom}(U, L)_0$$

where $U \subset \Delta[1] \times \Delta[n+1]$ is the smallest simplicial subset such that both $\Delta[n+1] \amalg \Delta[n+1]$ and $\Delta[1] \times \partial\Delta[n+1]$ map into it.

Proof. The first and third equalities are Simplicial, Lemma 017L. The second and fourth follow from the cited lemma combined with Simplicial, Lemma 018V. The last assertion follows from the fact that U is the push-out of the bottom and right arrow of the diagram (0.1.2), via Simplicial, Lemma 017N. To see that U is equal to this push-out it suffices to see that the intersection of $\Delta[n+1] \amalg \Delta[n+1]$ and $\Delta[1] \times \partial\Delta[n+1]$ in $\Delta[1] \times \Delta[n+1]$ is equal to $\partial\Delta[n+1] \amalg \partial\Delta[n+1]$. This we leave to the reader. □